



# Transformed objects

## Task A: Which transformation?

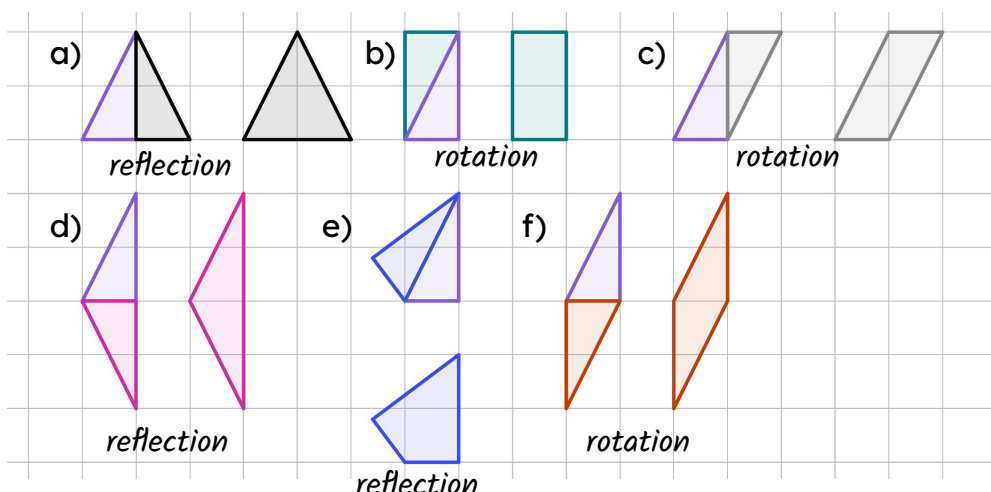
1) By considering the changes, decide which transformation could have taken place in each case. There might be more than one.

- |                                |                               |
|--------------------------------|-------------------------------|
| a) the lengths have changed    | <i>enlargement</i>            |
| b) the sense has changed       | <i>reflection</i>             |
| c) the orientation has changed | <i>reflection or rotation</i> |
| d) the position has changed    | <i>all</i>                    |

2) By considering the invariant properties, decide which transformation could not have taken place in each case. There might be more than one.

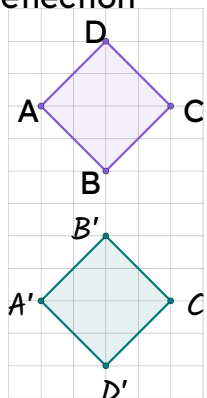
- |                                 |                                                                                        |
|---------------------------------|----------------------------------------------------------------------------------------|
| a) the lengths are invariant    | <i>not an enlargement</i>                                                              |
| b) the sense is invariant       | <i>not a reflection</i>                                                                |
| c) the orientation is invariant | <i>neither reflection nor rotation</i>                                                 |
| d) the angles are invariant     | <i>none (since all 4 transformations you are familiar with have invariant angles!)</i> |

3) Identify the type of transformation that has taken place to create the new shape.

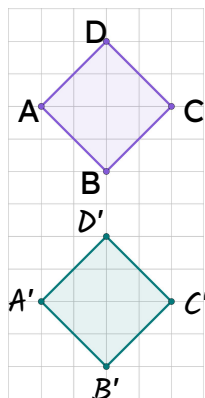


4) Add the image vertices labels.

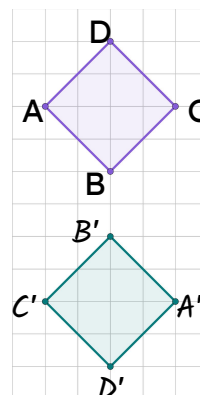
a) After a vertical reflection



b) After a translation



c) After a 180° rotation



## Task B: Recognising and describing transformations

1) Complete the descriptions.

a) Shape A has been reflected in a diagonal line passing through  $(-3, 3)$  and  $(1, -1)$  onto shape D.

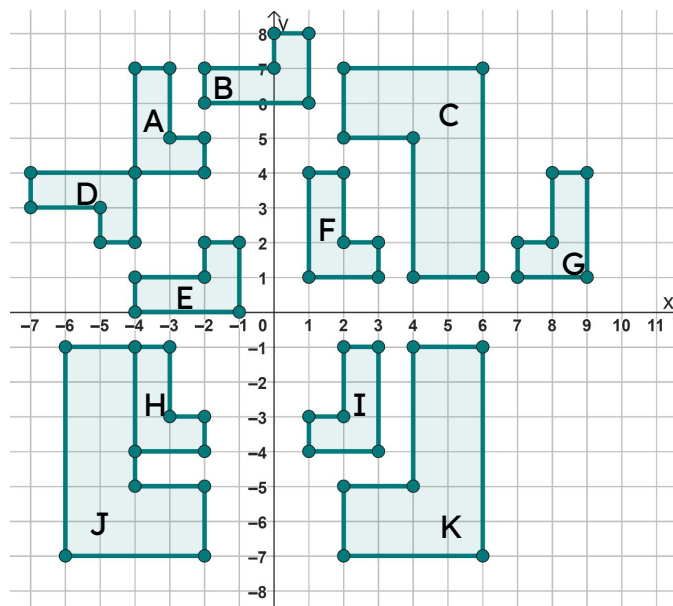
b) Shape J has been reflected in the y-axis onto shape K.

c) Shape B has translated by  $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$  onto shape E.

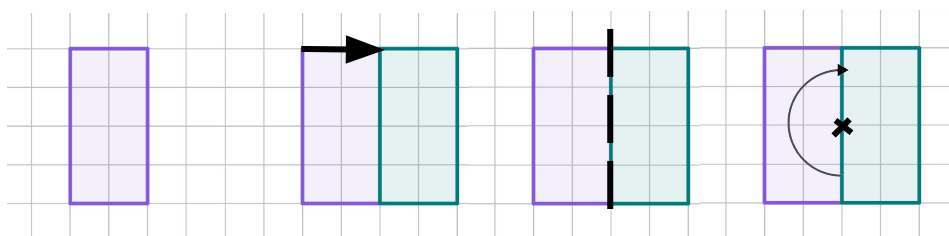
d) Shape H has been enlarged by scale factor 2 from  $(-2, -1)$  onto shape J.

e) Shape H has translated by  $\begin{pmatrix} 0 \\ 8 \end{pmatrix}$  onto shape A.

f) Shape C has been rotated 180° about the origin onto shape J.



2) Three different transformations have taken place to create the image, which combined makes a square. What are they?



1) Translate by  $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

2) Reflect along the right edge of the shape.

3) Rotate 180° around the midpoint of the right edge.

